

R.O.B. VISION

VIRTUAL ROBOT COMPANION / MK.01



USER GUIDE

Play, preview, setup, and problem solving

VERSION 0.1.7

UNO Q · RETROPIE · BATOCERA

CHAPTER 01 / STORY

Meet Buddy

Buddy lived in the dark between game frames. A flash told him to turn; another told him to lift. He could move the pieces, but the edge of the screen was as far as he had ever gone.



Then a little grid of lights appeared beyond the game. Buddy blinked back. The lights became eyes on the UNO Q, and a larger window opened. For the first time, he could see his hands, the game table, and someone waiting for him to move.

Now each signal gives Buddy another step into the room. When the game ends, he waits for the next round.

CHAPTER 02 / USER GUIDE

R.O.B. Vision user guide

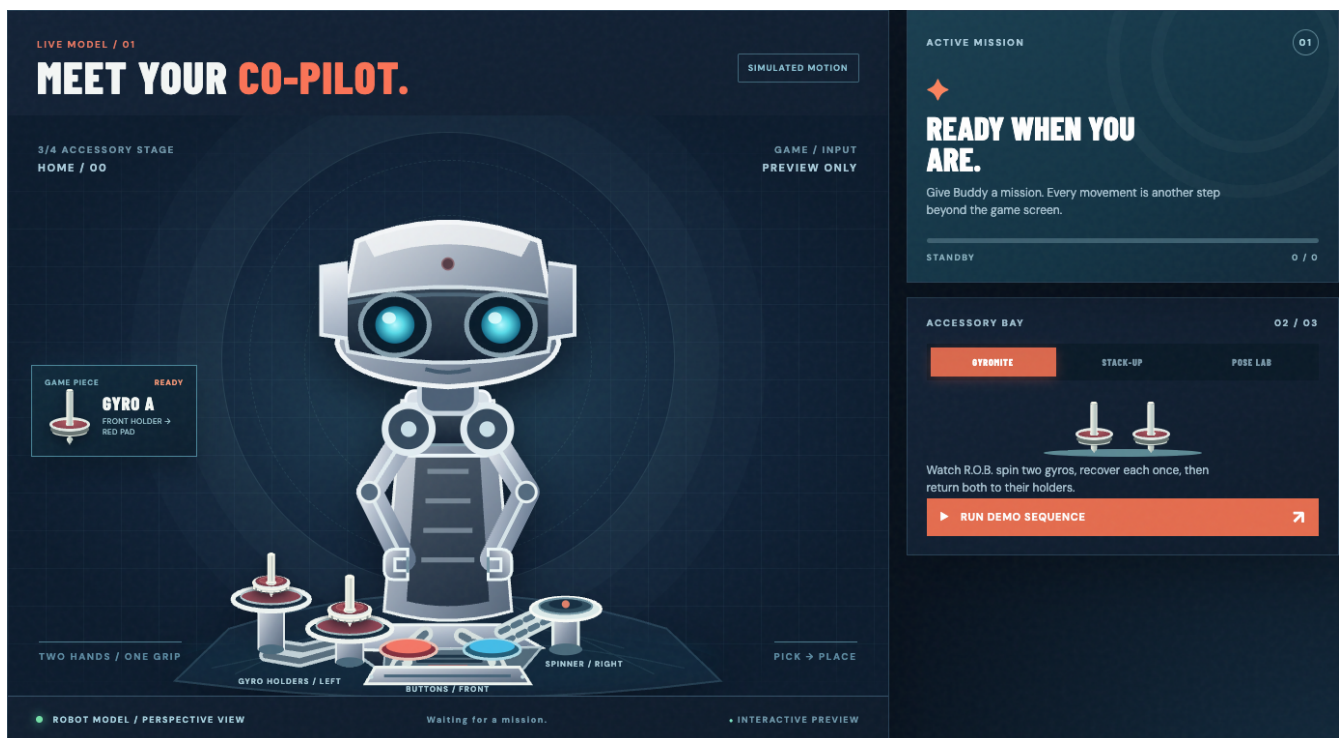
The UNO Q controls Buddy, the virtual robot shown in Mission. RetroPie or Batocera sends Gyromite and Stack-Up commands through the game-frame link. His movements and game pieces appear in your browser.

Open R.O.B. Vision

Install version 0.1.7 with the two commands in the [installation guide](#): one on the UNO Q, one on the console. The console command prints first-time pairing details for Setup and configures its receiver automatically.

Start **R.O.B. Vision** in UNO Q App Lab, then open one of the Mission URLs printed by the installer on a laptop, phone, or tablet. The installer prints both the UNO Q's `.local` hostname and its LAN IP address. Stop another App Lab app first if one is running. A `file://` page is an offline preview.

Mission shows R.O.B., Accessory Bay, Game Table, Pose Preview, System Vitals, and Activity Feed. **GYROMITE / CONNECTED** means the browser reaches the UNO Q with Gyromite selected. **RETROPIE ONLINE** or **BATOCERA ONLINE** means that console's receiver has contacted the UNO Q recently. **GAME FRAMES LINKED** means the selected game is sending its rendered frames.



Try a demonstration

When no supported game is running, select **Gyromite** or **Stack-Up** and **Run Demo Sequence**. Gyromite demonstrates a held unspun gyro, two spinning gyros on pads, spin-down, re-spin, and return to holders. The illustrative spin clock is 55 seconds. Stack-Up moves red alone and then a blue/white group. **Home** resets the pieces. Pose Lab is a character-only movement sandbox. A live game launch stops the demo and selects that game's scene.

Choose an NES emulator

For automatic R.O.B. movement on RetroPie, select **lr-robvision-fceumm** or **lr-robvision-nestopia** for the game. Each entry uses the corresponding installed core and forwards the game's rendered light cells to the UNO Q. On supported Batocera, the installer selects the R.O.B. Vision FCEUmm wrapper for registered games; Nestopia is available as a per-game choice. See the [RetroPie](#) and [Batocera](#) setup guides.

Play Gyromite

Open [Setup](#) to check the console and game-frame status. Start Gyromite and watch Mission for **GAME FRAMES LINKED**. R.O.B. follows complete commands from the game. A spinning virtual gyro on a pad keeps its gate pressed while R.O.B. moves elsewhere; an unspun gyro can press one gate while held there. The UNO Q sends those pad states to the active console's virtual Controller 2.

Fast Gates responds immediately while Gyromite is selected. Tap **Lower Blue** or **Lower Red** to press, tap again to release, or choose **Release Both**. On Mission, **2** toggles blue, **1** toggles red, and **0** releases both while you are not typing in a field. Holds expire after 60 seconds; both buttons release on game exit or lost receiver data. Buddy's hands must be empty, the matching gyro must be on its holder, and the target pad must be free. Return the pieces with regular controls or choose **Home** to reset the gyros. Watch the game screen to confirm the game's response.

Play Stack-Up

The live model begins with five colored blocks on Tray 3. In Direct mode, jumping Hector onto a command tile sends **Left**, **Right**, **Up**, **Down**, **Open**, or **Close** through the frame link. Manual controls use the same virtual model. A lower grip carries the contacted block and all blocks above it. Impossible moves are rejected and leave pieces in place. **Home** restores the initial stack.

Current game limits

Stack-Up's virtual arrangement does not score a round or initialize Bingo's distinct starting layout. Use the game screen to check its target and result. Stack-Up does not use Gyromite's gate-button return path.

Test mode and reset

Launch the game through its R.O.B. Vision emulator choice and enter Test mode. The game-frame link makes R.O.B.'s red light blink automatically on Mission and Setup when it recognizes the Test signal. A separate ready-light command makes the light steady briefly. Exiting Test mode returns the UNO Q matrix to the game's eyes. Neither signal moves R.O.B. **Preview Red Light** on Setup demonstrates the animation without claiming a game signal.

Home resets the selected virtual game. Live **Emergency Stop** clears the game selection; in the offline preview it cancels the script until reset. A browser reload restores the UNO Q's current snapshot. See [Setup](#), [gameplay](#), and [troubleshooting](#).

CHAPTER 03 / USER GUIDE

R.O.B. Vision gameplay guide

The UNO Q updates Buddy and the game pieces shown in your browser. A paired RetroPie or supported Batocera console sends movement commands from the game; Gyromite's virtual pad states return to that console's Controller 2.

Gyromite

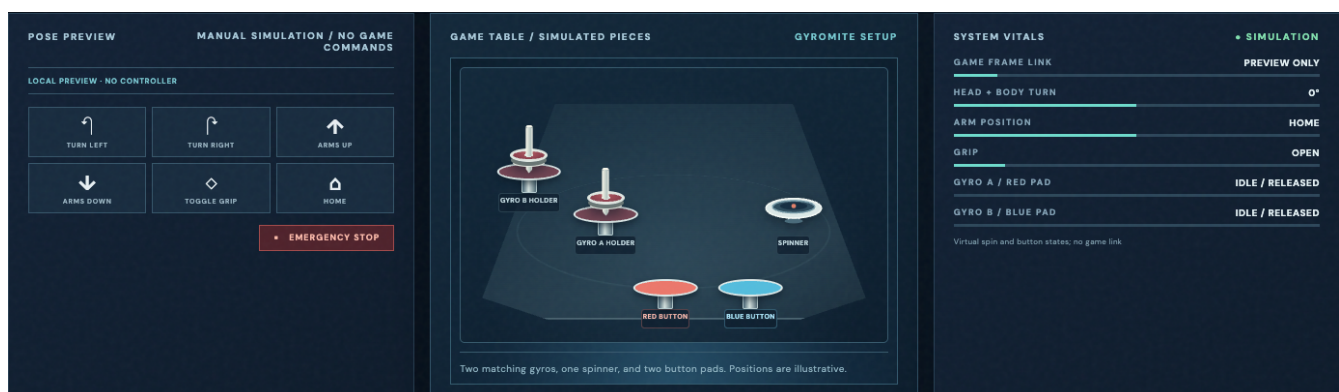
The [original Gyromite booklet](#) describes **Test**, **Direct**, **Game A**, and **Game B**. Direct sends six R.O.B. actions: up, down, left, right, open, and close. In Game A, Start switches into the robot-transmission screen for a command and then back to the professor. In Game B, the professor walks while Controller 1 directly commands R.O.B. A gyro on the blue or red tray lowers the matching gate.

The manual explicitly says a gyro **does not need to spin to operate only one gate**. R.O.B. can hold an unspun gyro on the virtual pad, then lift it to release that pad. If both gates must be controlled, spinning one gyro lets it stay upright on one pad while R.O.B. uses his hands to put the other on the other pad. A gyro that is no longer needed belongs on its holder. The spinner is a means to free R.O.B.'s hands, not an obligatory step for every pad press.

In Mission's live model, Buddy starts at the top (level 6). Two Down commands move through level 4 to the shared working height, level 2, where his hands meet either gyro holder, the spinner, or either colored pad. This is the virtual model's working height; the original manual's positions 1-5 label slots around the base, not arm levels. A gyro resting on a colored pad presses its button at level 2. Raise a carried gyro once to level 4 before turning; level 6 is also available for higher clearance. Their different screen positions come from the scene's 3D perspective, not different command heights.

Try the demo: with no game running, select Gyromite and **Run Demo Sequence**. Buddy presses a pad with an unspun gyro, uses two spinning gyros, and returns them to their holders. The sequence stops at **COMPLETE**. Select **Home** to reset. The demo's spin lifetime is illustrative.

During play: the UNO Q tracks both gyros and sends red and blue pad states to the active console. It releases both buttons on game exit or a lost link. **Fast Gates** presses or releases a matching pad immediately; each hold expires after 60 seconds. Watch the game screen for Hector and the gate response.



Pose Preview, Game Table, and System Vitals. Fast Gates appears during live Gyromite play.

Stack-Up

The [Stack-Up booklet](#) describes Direct, Memory, and Bingo. Buddy's Game Table has five trays and five colored blocks. Direct and Memory start with all five on Tray 3, top to bottom **red, white, blue, yellow, green**. In Direct mode, move Professor Hector onto the **left, right, up, down, open, or close** command keys. Memory plays a programmed command series; Bingo sends a command when one row or column completes. Buddy can carry a block with all the blocks above it.

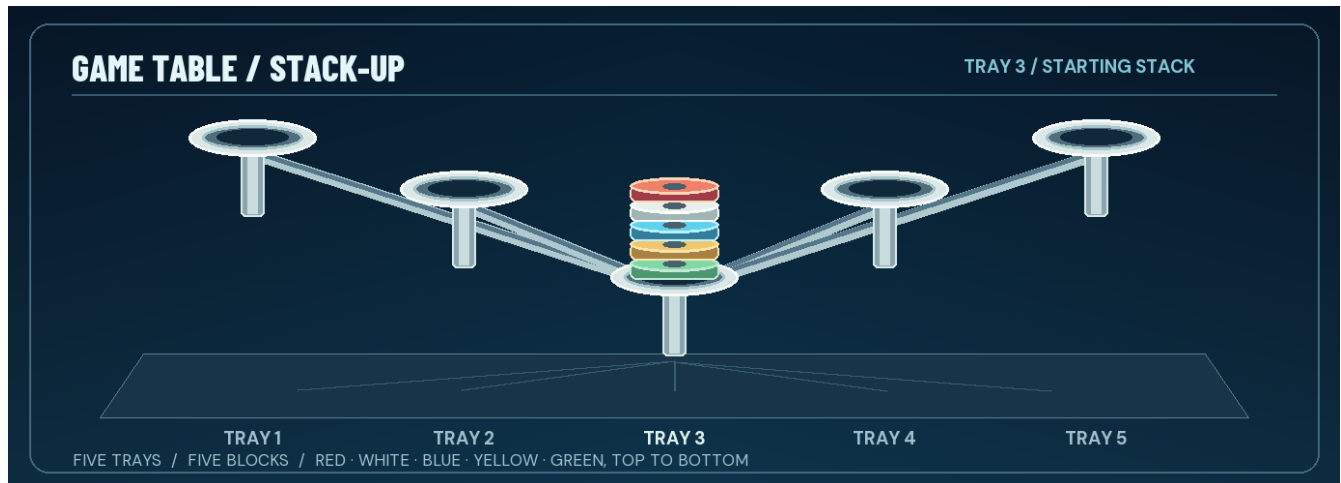


Illustration of the initial Stack-Up Game Table, using the dashboard's tray layout and block order.

Try the demo: with no game running, select Stack-Up and **Run Demo Sequence**. Buddy moves red alone, then carries blue and white together. All five blocks remain visible in the Live Model, Accessory Bay, and Game Table. The **Stack-Up Commands** buttons apply one virtual action at a time. **Home** restores the starting stack. If a move is blocked, the pieces stay in place and the Activity Feed explains why.

Current limits: Buddy starts with the Direct/Memory stack in every Stack-Up mode; Bingo's different starting arrangements are not modeled. Compare the virtual arrangement with the game's target on its screen. The game sends movement commands but does not tell Buddy its target, score, or whether the round was completed. Stack-Up does not use Gyromite's gate-button return path.

Pose Lab and recovery

Pose Lab lets you try Buddy's head, body turn, arms, and shared grip. Its short demo is a greeting. If the browser reconnects during a game, it resumes from the UNO Q's current state. If the Gyromite link drops, both virtual gate buttons release.

R.O.B. Vision Setup

Install the [current release](#) with one command on the UNO Q and one on RetroPie or Batocera. The console command detects its platform and opens first-time pairing automatically. The Batocera installer checks for an installed FCEUmm or Nestopia core and a usable native frame wrapper before changing files.

Open the Setup URL printed by the UNO Q installer: `http://<hostname>.local/dashboard/setup.html` or `http://<LAN-IP>/dashboard/setup.html`. Console Link lists each paired RetroPie or Batocera console with its address and live status. Then make these five checks:

1. **Pair the console.** Keep the console installer running while it displays a six-digit code and certificate fingerprint for five minutes. Enter its hostname, code, and fingerprint on Setup, then select **Pair Console**. The console command starts the receiver and configures Gyromite's virtual Controller 2 when pairing completes. Use **Pair Another Console** to add another. Each appears under **Paired Consoles**; **Check Link** refreshes the receiver link. **Receiver Waiting** means this UNO Q has not received a recent authenticated check-in. It does not say whether the console is powered on or whether its `.local` name resolves. Existing pairings remain saved when the UNO Q app is upgraded. A console uses one UNO Q at a time; pairing it with another updates the receiver address, while the old UNO Q retains a waiting row until you remove it.
2. **Register your ROM filenames.** Select **Edit ROMs** beside the paired console. This opens that console's **Game Registry** editor farther down Setup. The common World filenames are provided. For another filename, add its exact basename and extension under games with the value `gyromite` or `stack_up`, select **Validate**, then **Save to Console**. Exit any running game before saving. Each console has its own registry, retained through upgrades.
3. **Check game frames.** On RetroPie, select `lr-robvision-fceumm` or `lr-robvision-nestopia`. On Batocera, launch an exact registered ROM; its R.O.B. Vision FCEUmm wrapper is selected automatically. Setup should show **GAME FRAMES LINKED**. That means the game is sending frames R.O.B. Vision can use for movement and Test signals.
4. **Check Test mode.** Enter the game's Test mode. R.O.B.'s red light should blink automatically on Mission and Setup, and the UNO Q matrix should show a pulsing **T**. A separate ready-light command makes the light steady briefly. **Preview Red Light** demonstrates the animation when no Test signal is present. If the light stays off, check **GAME FRAMES LINKED** and confirm the game is in Test mode. In Stack-Up, press Select at the ROBOT BLOCK artwork to show the mode list, choose Test, then press Start.
5. **Try the controls.** For Gyromite, **Lower Red** and **Lower Blue** press matching virtual pads immediately; press again to release or choose **Release Both**. Each hold expires after 60 seconds. For Stack-Up, **Left, Right, Up, Down, Open, Close** send one command to the virtual model. Watch Mission's Live Model and Game Table for movement.

A file:// Setup page is only an offline preview. Use the UNO Q URL for live status. Mission keeps Fast Gates and the Game Table next to Pose Preview and System Vitals.

To forget a console, select **Remove** in its row and then **Confirm Remove**. The UNO Q immediately revokes that console's credential; any other paired console remains connected. If the removed console was playing, its game session is cleared. The removed console retains its old local credential, but it can no longer authenticate; rerun the one console install command with `--pair` at the end to reconnect it. Only one console can be the active game source at a time.

For physical gamepad assignments and Buddy's Player 2 route, see the [Controller Router Guide](#).

CHAPTER 05 / USER GUIDE

Controller Router

Controller Router decides which game controller supplies each player in RetroArch. R.O.B. Vision installs it on RetroPie and Batocera so Buddy's virtual buttons and your physical gamepads can coexist. Your console still uses its original controllers in EmulationStation; Router takes over their game input only while a Libretro game is running.

What goes to each player

Source	Usual assignment	What it does
Your physical gamepad	Player 1	Moves Hector, selects a mode, starts the game, and provides the console's normal menu and exit controls.
Buddy (R.O.B. Vision Controller 2)	Player 2	Presses Gyromite's red and blue gates when a gyro rests on a pad or you use Fast Gates. Buddy's visible Stack-Up movements come from decoded game frames.
Another physical gamepad	Any chosen player	Can share a player with another pad or control a different player if the game supports it.

Buddy stays on Player 2. On a new Router installation, gamepads already configured in EmulationStation are suggested in Player 1–4 order. If Router was already installed with VirtualGlove, R.O.B. Vision keeps its saved assignments and adds Buddy to Player 2. A controller's Linux jsn number and RetroArch pad number can change after reboot; use the named assignments in Setup instead of editing those numbers.

Router assigns controller sources to player slots. It does not alter an NES game's player count, the ROM registry, the game-frame link, or the buttons you mapped in EmulationStation. A physical controller may share Buddy's Player 2 slot, but it will then send Player 2 input during Libretro play.

Review or change assignments

1. Open the live UNO Q **Setup** page and find **Console Link**.
2. Choose **Controllers** beside the console you are playing. The Controller Router card selects that console and shows its saved assignments.
3. Check which physical gamepad is Player 1. If you use more than one console, review each separately; each console saves its own Router settings.
4. Choose **Test Inputs**, then press a direction or button on each connected gamepad during the five-second test. The result tells you which controls were seen. A pad can be connected yet show no activity if you did not press it during the test.
5. To change an assignment, exit the RetroArch game, choose the desired player beside that physical gamepad, then select **Save Assignments**. Relaunch the game to use the new routing.

Reload fetches the current console assignments and discards unsaved choices on the page. **Restore Previous** returns to the last saved assignment; it is available only when a previous save exists. Buddy's Player 2 choice cannot be changed in R.O.B. Vision Setup.

What the status means

- **Connected:** Router can see that configured physical controller now.
- **Unavailable:** A saved controller is not present or its identity no longer matches. Reconnect the same pad. If you replaced it, configure the replacement in EmulationStation, then assign it in Router.
- **Mapping updated:** EmulationStation has a newer valid button map for the same controller. Router keeps its player assignment and uses the refreshed map at the next game launch.
- **No input detected:** No button or direction was pressed during the test. Try again while moving the pad; this result alone does not prove the pad is broken.

The merged Player 1–4 devices stay neutral in EmulationStation. Test the original gamepad there, then test its merged player inside a Libretro game.

When a controller seems wrong

My gamepad works in EmulationStation but not in the game. Try another connected gamepad first. The one in your hands may be Player 2 while another is Player 1. Review the names and assignments in Setup, use **Test Inputs**, and correct the slot with the game closed.

Hector moves but Gyromite's gates do not. Player 1 working does not prove Buddy's Player 2 path. Check that Buddy appears on Player 2, the console link is online, and Gyromite is running through a R.O.B. Vision FCEUmm or Nestopia choice. On the game screen, test one gate at a time with **Lower Blue** or **Lower Red** in Setup, then choose **Release Both**.

Buddy moves on Mission but a physical pad does nothing. Confirm that the pad is assigned to the player you intend, then check its activity. If it was remapped in EmulationStation, close and relaunch the game. Do not change RetroArch's generated joystick index to chase a changing device number.

A controller disappeared after it was unplugged. Reconnect the saved pad and reload the card. Router releases held input from a disconnected source. If the old pad is gone permanently, assign its replacement and save while no game is running.

For installation and pairing, see [Installation and Setup](#). For game-specific checks, see the [Game Manual](#) and [Troubleshooting](#).

R.O.B. Vision installation guide - 0.1.7

R.O.B. Vision runs on an Arduino UNO Q and connects to either RetroPie or Batocera. Install the UNO Q first, then install each console. **Each device has one command for both installation and upgrades.** Rerun that device's command to get the latest release; there is no separate upgrade command. The release installer downloads and verifies the app before installing it.

Before you begin

- Connect the UNO Q and console to the same trusted network. Sign in to the UNO Q as `arduino`, to RetroPie as `pi`, or to Batocera as `root`.
- Install UNO Q App Lab. Standard RetroPie needs Python 3.7+, `gcc`, `openssl`, `sudo`, `systemd`, `uinput`, and at least one installed NES core (`lr-fceumm` or `lr-nestopia`). The UNO Q needs Python 3.9+. Batocera needs Python 3.9+, its service manager, SDL2, and an installed FCEUmm or Nestopia core. The installer uses a matching packaged frame wrapper or compiles one when a native C compiler is available.
- Exit any running game and EmulationStation on RetroPie. Restarting the virtual joystick while EmulationStation is open can crash its input manager. Stop a different App Lab app before installing R.O.B. Vision. Supply your own legally obtained Gyromite or Stack-Up ROM. After pairing, select **Edit ROMs** beside that console in Setup to add a different filename to its registry.
- `curl`, `tar`, Python 3, and either `sha256sum` or `shasum` are needed to run the single-command download. The installer reports a missing prerequisite before installing.

Install or upgrade the UNO Q

Run this one command in the UNO Q terminal as `arduino`:

```
curl -fsSL https://github.com/mathan416/ROB-Vision/releases/latest/download/install.sh | sh -s -- -uno-q
```

Use this exact command again to upgrade or repair the UNO Q. It downloads and verifies the latest R.O.B. Vision release, installs the App Lab app, and starts it. On upgrades, App Lab stops and restarts the app while preserving its pairing data. App Lab also compiles and uploads the UNO Q matrix sketch. At the end, the installer prints Mission and Setup links using the device's `.local` name and available LAN IPv4 addresses. Open either form in a browser on the same network.

Install or upgrade a console

Run this one command **on RetroPie or Batocera**, replacing your `-uno-q.local` with your UNO Q's LAN hostname or IP address:

```
curl -fsSL https://github.com/mathan416/ROB-Vision/releases/latest/download/install.sh | \
sh -s -- -console your-uno-q.local
```

Use this exact command again to upgrade or repair that console. It detects RetroPie or Batocera. On RetroPie it uses `sudo` for system changes and requires EmulationStation to be closed. On Batocera it runs under the root login and temporarily suspends and resumes its game menu while the virtual joystick is restarted. It installs the frame wrappers, receiver, game launch integration, and Controller Router with Buddy assigned to Player 2. For a first install it opens a five-minute pairing window and prints a six-digit code and SHA-256 certificate fingerprint. Enter the console's `.local` hostname or LAN IP address, code, and fingerprint under **Console Link** on the UNO Q Setup

page. Keep the console command open until it reports **Pairing complete**. Successful pairing starts the receiver, installs or updates Controller Router, and assigns Buddy to Player 2. Existing physical-controller assignments remain in place. No second terminal command is required.

Upgrades and re-pairing

An upgrade retains that console's credential and restarts its receiver. Rerun the command in **Install or upgrade a console** for a repair or upgrade. If you remove a console in Setup, rerun the console command with `--pair` added at the end to open a new pairing window; this is still one command on the console.

The installer leaves ROM files and unrelated games alone. RetroPie registers `lr-robvision-fceumm` and `lr-robvision-nestopia` for the exact registered ROM names. Batocera selects its R.O.B. Vision FCEUmm wrapper for those names while retaining unrelated core choices. Plain NES cores still play games but do not send commands to R.O.B. Vision.

Review controller assignments

On the UNO Q Setup page, choose **Controllers** beside a paired console. Buddy stays on Player 2. The list shows physical controllers configured in EmulationStation and their current player assignments. On a new Router installation, known controllers are seeded in Player 1–4 order. Change an assignment if needed, exit the running game, and choose **Save Assignments**. **Test Inputs** listens briefly for buttons and directions. **Restore Previous** returns to the last saved assignment. RetroPie requires EmulationStation closed during first pairing or Router installation because new virtual controllers can upset its input manager.

Check the link and play

On Setup, select **Check Link** after pairing. It should show the console online. Launch Gyromite or Stack-Up using a R.O.B. Vision core; Mission and Setup should show **GAME FRAMES LINKED**. In Gyromite Test mode the Test light blinks automatically. In Game A, test blue and red gates independently. Stack-Up Direct mode supplies Left, Right, Up, Down, Open, and Close. The virtual robot and game pieces appear in Mission on a laptop, phone, or tablet.

If a game is already running during installation, the console installer stops before changing its configuration. Exit the game and run the same command again. On Batocera, missing NES cores or a usable native wrapper are reported before any changes. For a failed link, see [Troubleshooting](#) and the built-in Help page.

Supported releases and integrity

The Batocera installer checks installed NES cores and a loadable frame wrapper rather than a fixed board or version. The package includes FCEUmm and Nestopia wrappers for x86_64, 32-bit x86, AArch64, ARMv7, ARMv6, and RISC-V 64. If a packaged wrapper cannot load, a native C compiler can build one before installation changes the device. Live device testing covered Batocera 43.1 x86_64; other architectures receive the same preflight checks but have not all been tested on physical hardware. Release assets include the source package, installer, PDFs, and SHA256SUMS. The installer checks the source package's digest before installing it.

For advanced repair, see the [RetroPie](#) and [Batocera](#) guides.

CHAPTER 07 / USER GUIDE

R.O.B. Vision troubleshooting

If RetroPie's EmulationStation exits with an InputManager joystick assertion, return to its shell and rerun the console installer with EmulationStation closed. The installer checks module compatibility before restarting the virtual controller.

For installation failures, check the prerequisite or unsupported-device message from the single release command, then rerun that same command. The archive checksum must pass before the installer changes the device. After removing a console in Setup, rerun its command with `--pair`.

First check which page is open: use the UNO Q installer's device-specific `.local` or LAN IP dashboard link for live state. A `file://` page or a simple static web server shows the independent preview. If `.local` does not resolve on a device, use the printed IP link on the same LAN and check Avahi on the UNO Q.

Symptom	Check
App does not start, port 8766 in use	Stop VirtualGlove or the separate <code>rob-vision.service</code> ; App Lab and that service cannot both bind the port.
GYROMITE / CONNECTED but no robot movement	Check the active console's ONLINE indicator and GAME FRAMES LINKED separately. If the latter is absent, restart the game with its R.O.B. Vision core. Check Mission's Activity Feed for decoded or blocked actions.
Console is online but no game is selected	A receiver can be online with no active game. Launch a registered ROM; then reload Mission if needed.
Test light never blinks	Enter the game's Test mode and confirm GAME FRAMES LINKED with the correct game selected. The light responds automatically to linked game frames. A red-light preview is only an animation demonstration.
Movement command is missed	Confirm GAME FRAMES LINKED and the correct game is selected. On RetroPie, choose <code>lr-robvision-fceumm</code> or <code>lr-robvision-nestopia</code> ; on Batocera, launch the registered ROM with its installed R.O.B. Vision wrapper. Check the Activity Feed for a decoded command or a model move blocked by height, grip, or station.
Red/blue gate does not move	Confirm Gyromite is the active RetroArch game, Setup Check Link is online, and RetroArch port 2 is configured. Inspect the game screen during an isolated color hold.
A physical gamepad does not control the game	Try another connected gamepad. The one in your hands may be assigned to Player 2 while another is Player 1. In Setup > Controller Router, check the assignments, move the intended pad to the right player if needed, and relaunch the game.
Fast Gates vanished	They show only in live Gyromite.
Gate remains down	Select Release Both ; a hold also expires at 60 seconds. The receiver releases on game exit or stale network data.
Stack-Up block does not move	Review current station, height, gripper, carried group, and destination capacity; invalid moves leave the stack intact and appear in activity.
Demo does not start	Exit the running game first. Demo mode works when paired and idle; Home resets it.
Browser shows stale scene	Reload for a fresh <code>/api/state</code> snapshot. A local preview cannot show UNO Q state.

On RetroPie, check `rob-vision-controller2.service`, `runcommand`, and the ROM basename. On Batocera, check `ROBvision`, the per-ROM core, and the ROM basename. Linux `/dev/input/jsN` may differ from RetroArch's pad index. Check Setup > Controller Router for the saved player assignment; Router resolves the active pad indexes. Exclude credentials and ROM files from bug reports.